

UBT517: BIOPROCESS ENGINEERING

L	T	P	Cr
3	1	0	3.5

Course Objective: The objective of the course is to apply fundamental principles and concepts of chemical engineering to biological systems. This course will provide a comprehensive understanding of media formulations, microbial growth kinetics, bioreactor selection, upstream & fermentation processes, and its role in manufacturing bio-products.

Introduction: Introduction to bioprocess engineering, Interaction of two disciplines: biotechnology and bioprocess engineering, outline of unit operations involved in upstream and downstream processing, steps in bioprocess development, role of bioprocess engineer in the biotechnology industry, historical advancement in bioprocess technology, range of microbial diversity, process and products relating to bioprocess engineering.

Microbial Growth Kinetics: Growth, growth measurement, media formulation, simple and complex media, media sterilization, Del factor, design of batch and continuous sterilization, log penetration theory, scale up of sterilization process, filter design, screening, culturing, culture preservation, strain improvement, inoculum preparation, stoichiometry of cell growth and product formation, elemental balances- degrees of reduction, factors influencing product formation on varying carbon & nitrogen source, batch culture, Monod's kinetics, modelling of batch growth kinetics, environmental factors affecting microbial growth, continuous culture, an ideal chemostat, advantages and limitations of continuous over batch culture, fed-batch culture and its applications.

Bioreactor Selection, Design and Process Control: Overview of reactor, mass balance in bioreactor, design equations of bioreactors, selection criteria for bioreactor, body construction of fermenter and its components i.e., impellers, stirred glands and bearings, seal assemblies, baffles, sparger and valves, solid state and submerged fermentation, design aspects of bubble column bioreactor, air-lift fermenter, plug-flow and packed bed bioreactor, scaling up of bioreactor. Main parameters to be monitored and controlled in fermentation processes, control systems in a bioprocess, methods of measuring process variables i.e., temperature, pressure, flow, dissolved oxygen, pH, role of computers in fermentation process analysis.

Transport Phenomenon in Bioreactor: Fick's law, theories of mass transfer, mass transfer between two phases, role of aeration and agitation in a bioprocess, oxygen transfer methodology in a fermentation process, significance of volumetric transfer coefficient (KL_a) and its determination, factors affecting KL_a values in a bioreactor, power requirements in gassed and un-gassed bioreactors.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Course Learning Objectives (CLO)

The students will be able to:

1. Explain how microorganisms and biochemical processes can be applied in engineered systems.
2. Distinguish among batch, continuous and fed-batch culture systems for the production of biochemical products.
3. Describe microbial growth & cultivation, various bioreactor components, and types of bioreactors used in biotechnology industries
4. Design media sterilization and design of air filter in a bioprocess.
5. Apply various concepts to improve bioreactor performance and evaluate process variables to analyse a bioprocess.

Text Books

1. Stanbury PF, Hall SJ, and Whitaker A, Principles of Fermentation Technology, 3rd Edition, Elsevier (2016).
2. Shuler ML and Kargi F, Bioprocess Engineering, 2nd Edition Prentice Hall (2015).
3. Sivasankar B, Bioseparations: Principles and Techniques, PHI Learning Pvt. Ltd. (2006).
4. Belter PA, Cussler E and Hu WS, Bioseparation – Downstream Processing for Biotechnology, Wiley Interscience (1988)

Reference Books

1. Doran P M, Bioprocess Engineering Principles, 2nd Edition, Academic Press (2012).
2. Ahuja S, Handbook of Bioseparations, Academic Press (2000).
3. Harrison RG, Bioseparations: Science and Engineering, Oxford University Press (2015).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-45
3	Sessional (May include Assignments/Projects/Tutorials/Quizzes)	30

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